

Co-pyrolysis of biomass and plastic wastes and application of machine learning for modelling of the process: A comprehensive review

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ABSTRACT

The conventional fossil fuels which primarily include coal, oil and natural gas are the major source of greenhouse gas emissions (such as methane, carbon dioxide and nitrous oxide) into the atmosphere causing severe health consequences to human population. Different types of renewable energy feedstocks including biomass wastes are being investigated across the world. Out of various techniques for utilizing biomass, the pyrolysis has wide product profiles which can be used in different applications. Likewise, omnipresence of plastic waste, and its tremendous generation and lack of appropriate waste management system is also another environmental issue. Hence, co-pyrolysis (a thermochemical conversion) of biomass and plastic waste, presents an effective solution for the underlined issues as it not only provides a clean source of energy, but is also cost-efficient, easy to use, helps deal with the issue of plastic waste management as well as mitigate the concerns caused by the pyrolysis of single feedstock i.e., biomass. The quality of co-pyrolysis derived bio-oil can further be enhanced by incorporating catalyst. Operating condition of a pyrolysis process depends on the nature of feedstock, requirement of product distribution etc. Thus, optimization of process parameters is essential for making this process successful. Machine learning models can be utilized in the co-pyrolysis process as a tool to overcome the preceding issues by optimizing the process and also helps in process control, yield prediction and real-time monitoring. However, no prior study has conducted an in-depth review of current research scenario related to the machine learning approach in co-pyrolysis process.

1. Introduction

The fossil fuel (coal, petroleum, and natural gas) demand is increasing day by day due to the tremendous growth of human population and industrialization in the world. Due to the usage of massive amounts of fossil fuels, greenhouse gas (GHG) emissions are increasing in the environment causing global warming and simultaneously affecting human health, causing sharp and undesirable climate changes and also leading to depletion of non-renewable energy sources [1]. Therefore, the solution of these severe problems is to find an alternative fuel that can be environment friendly and renewable in nature.

Biomass is emerging as a major source of renewable energy and is also categorized as a carbon-neutral fuel because the carbon dioxide (CO₂) released after it is burned is roughly equivalent to the CO₂ absorbed by the plants during the process of photosynthesis. Biomass includes trees, plants, crops, and animal waste etc. [2,3], and the bio-polymers present in biomass are cellulose, hemicellulose and lignin

[4,5]. Through incorporation of biochemical or thermochemical methods, biomass can be used to get subsequent bio-oil, bio-char, NCGs (non-condensable gases) and other carbon rich energy products/fuels [5,6]. Biochemical conversion includes fermentation and anaerobic digestion, while thermochemical conversion methods includes combustion, hydrothermal liquefaction, pyrolysis and gasification. Thermochemical conversion has the higher conversion rate in comparison to biochemical conversions due to the complete chemical nature of reaction, fast reaction rate, and production of many valuable carbon rich byproducts. Thermochemical methods are very effective for conversion of certain type of wastes, like plastics, which cannot be degraded by microbes.

Pyrolysis, gasification, liquefaction, and combustion are the commonly used thermochemical techniques to extract energy from biomass [6–9]. In case of pyrolysis, there is a thermal decomposition of feedstocks (having less moisture) at an elevated temperature and very less time in the presence of an inert atmosphere (in the presence of

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nitrogen, helium and argon) [10–12]. Different oxygenated compounds such as carboxylic acids, aldehydes, alcohols, lignin-derived oligomers, carbohydrates, etc. are present in the pyrolysis derived bio-oil [6]. Biochar produced can also be utilized as an adsorbent, catalyst, fertilizer, etc. [13,14] and it also improves the quality of soil and also helps in carbon sequestration [15]. Pyrolysis is getting positive attention because to its good efficiency as well as sustainability [16,17]. Many ongoing research and studies are going on in the field of pyrolysis to make it more efficient and cost-effective. Although pyrolysis derived bio-oil is potent to be utilized as a sustainable and renewable fuel however, it needs to be upgraded in order to increase its competitiveness in terms of commercial range fuels [18] as pyrolyzed oil possess high moisture and oxygen contents (about 35–60 wt%), is unstable, corrosive, and viscous in nature, as well as also have low calorific value [18,19]. The higher heating value of pyrolyzed oil lies between 18 and 28 MJ/kg [20–22]. Therefore, upgradation of bio-oil is very necessary through new thermal processing and/or pretreatment method [23]. There are different upgradation methods for bio-oil e.g., hydrodeoxygenation (HDO), catalyst based pyrolysis, catalytic cracking, and molecular distillation etc. [15].

Plastic waste is ubiquitous in nature and till date, around 8 billion metric tons of plastic wastes have been produced [24]. Above 300 million tons of waste plastics are reportedly manufactured annually throughout the world, out of which 8 million tons of plastic enters the ocean every single year as pollutants [25,26]. The accumulation of plastic wastes can produce many unfavorable effects on the environment, as the discarded plastic can take up to hundreds of years to degrade on earth [26]. The surface and groundwater also gets polluted by the leaching of harmful chemicals from plastic waste [27]. However, it's not easy to recycle the plastic waste due to the presence of product selective additives in it as well as incineration of plastic waste also causes emission of harmful gases like SO_x, NO_x and other greenhouse gases which cause harm to wildlife, pollute water etc.

Co-pyrolysis is a process in which two or more materials are thermally decomposed together in the same reactor, producing a mixture of gases, liquids, and solids [28]. Co-feeding of plastics with biomass in the reactor helps to give better stability and higher heating value to the bio-oil as it reduces the amount of oxygen content present in the bio-oil [29,30]. Hydrogen transfer occurs during co-pyrolysis process of biomass and plastics wastes. The utilization of waste plastics through this process will not only minimize the accumulation of plastic waste in municipal solid waste (MSW) and landfills but will also promote awareness regarding the importance of plastic waste recycling. Moreover, it has the potential to mitigate environmental pollution while serving as a viable alternative energy source [31].

Further upgradation of the co-pyrolysis process can be achieved by the addition of catalyst as it promotes the decomposition of the materials and the formation of new selective products [14,32]. The catalyst can also help reduce the temperature and time requirement of the co-pyrolysis process, thereby reducing the energy and cost input [32].

The added advantage of catalytic co-pyrolysis over co-pyrolysis is that the reaction can be designed to yield particular fuels and aromatics since most of reactions happen at the catalyst site [29]. Catalytic co-pyrolysis of biomass and waste plastics also helps to remediate the issue of high viscosity, high acidity, comparatively low HHV (higher heating value) and chemical instability of bio-oil [1]. Xue et al. (2015) worked on co-pyrolysis of red oak and plastic wastes and discovered that yield of hydrocarbon increased synergistically during the catalytic co-pyrolysis of carbohydrates and PE in the presence of zeolite catalyst [33]. Likewise, According to Li et al. (2014), aromatics were produced during catalytic co-pyrolysis of furans with LDPE in the presence of ZSM-5 catalyst [34]. Hence, the catalysts are chosen based on their particular application and desired outcomes.

Further successful implementation of a process requires in-depth understanding of its kinetics and fundamentals, as well as comprehensive knowledge of optimized process parameters, which requires extensive experimentation, and calculations that leads to high labor

intensity and high operational cost. Hence, recent research trend is tilting towards application of machine learning (ML), an artificial intelligence tool that can help reduce the labor and cost intensity for selecting a process along with providing an exhaustive knowledge of the process fundamentals. ML approach can optimize the process, real-time monitoring and control of the process, for the analysis of process kinetics, product analysis and lifecycle analysis, etc. [35]. Artificial neural network (ANN), support vector regression (SVR), decision tree, and random forest (RF) are machine learning techniques that have recently been used in the co-pyrolysis of biomass to predict the kinetics as well as the optimal yield of the process [36].

Various research studies have been done for the modelling of the co-pyrolysis of biomass and plastic wastes through machine learning. Prasertpong et al. (2023) applied XGBoost (Extreme Gradient Boosting) machine learning model on co-pyrolysis of biomass and plastics using 26 input parameters [18]. Similarly, in a study by Cheng et al. (2023), DT (Decision Tree) and ANN (Artificial Neural Network) gave best R² values of 0.984 and 0.926 respectively for prediction of yield of bio-oil from pyrolysis of waste plastics [37]. Alabdrabalnabi et al. (2022) found XGBoost and neural network as an appropriate models for the modelling char and bio-oil yield during the biomass and plastics co-pyrolysis [38]. Multiple research work related to the co-pyrolysis of biomass and plastic waste were observed and scrutinized from the literature, however no study has conducted the in-depth analysis pertaining to the approach of ML algorithms in the process of co-pyrolysis of biomass and plastic wastes. Hence, current research scenario highlighting the utilization of machine learning for process optimization and in-depth study of a process mechanism is essential to identify the future scope of machine learning in association with the catalytic co-pyrolysis.

In recent years, a lot of research and review articles have been published on the co-pyrolysis of biomass and plastic waste. Zulkafli et al. [39] reviewed the influence of plastics (High density polyethylene (HDPE), polypropylene (PP), etc.) and various catalysts (Metal modified zeolite, mesoporous zeolite etc.) on the co-pyrolysis of biomass and plastics for producing renewable fuels. Raj et al. [40] conducted a study on the effect of metal impregnated catalysts on the kinetics of pyrolysis of biomass. Ryu et al. [41] reviewed the catalytic co-pyrolysis of biomass and plastic wastes for the production of hydrocarbon rich fuel. Hassan et al. [23] reviewed the catalytic co-pyrolysis of biomass with plastics and coal, highlighting that acid catalysts enhance the yield and selectivity of high-value added chemicals. Esso et al. [42] reported the review on the synergistic effects observed during the co-pyrolysis of biomass and plastic waste. Earlier reported studies were primarily focused on co-pyrolysis of biomass and plastics. Thus, there is significant gap in comprehensive studies on the application of ML for modelling of co-pyrolysis of biomass and plastic wastes. Consequently, this review paper aims to provide a clear overview of the current advancement in this field.

Likewise, till date no prior study has conducted a critical and comprehensive review on the application of machine learning in co-pyrolysis of biomass and waste plastics incorporating the catalytic effect on the upgradation of the product yield as well as spotlighted the feasible use of machine learning in various aspects of a process functioning. Thus, this review paper conducts an in-depth review pertaining to the co-pyrolysis of biomass and plastics, highlighting the process mechanism and current research scenario related to the catalytic co-pyrolysis. Further detailed review of the machine learning process and its current application in the field of pyrolysis and co-pyrolysis has been spotlighted along with the future scope recommendations.

2. Mechanistic study of co-pyrolysis

2.1. Co-pyrolysis (without catalyst)

Co-pyrolysis is a process of simultaneous pyrolysis of two or more feedstocks. The mechanism of co-pyrolysis involves the thermal

decomposition of the feedstock at elevated temperatures (300–600 °C) in the anaerobic condition. The bio-oil is one of the primary product of pyrolysis process [43]. It is an innovative method for improving the quality of oil produced through pyrolysis [44]. Synergistic effect is the main driver of improving the quality parameters and yield of bio-oil [15]. The main parameters for achieving the synergistic effect are the interaction between individual co-feeding materials (e.g., biomass and plastic waste) and their operating temperature, feedstock ratio, and reaction time. Effective H/C ratio of the feedstock, and feedstock type (biomass, plastics etc.), shape and size of feedstock are also the main parameters on which the quantity and quality of the bio-oil depends [5, 45].

A synergistic effect takes place due to interaction among biomass and plastic wastes during the co-pyrolysis process, wherein co-feeding of hydrogen-rich material (such as plastics waste, tires) with hydrogen-deficient biomass shows an increase in H/C_{eff} of overall feedstock [5, 15]. Co-pyrolysis is a process that involves initial breakdown, secondary reactions, homogenization, and the production of the final product [46]. At a specific temperature, the feedstock starts to break down into smaller molecules. More complex compounds are formed where the smaller molecules form the first breakdown undergo further events such as isomerization, condensation, and cracking [47]. The complex molecules derived by the secondary reactions interact with each other and become homogenized, followed by the formation of new compounds. These compounds undergo further reactions and rearrangements, leading to the formation of final products such as gases, liquids, or solids. The resulting products have unique properties and applications compared to the starting materials, making co-pyrolysis an important process in various industries such as energy, materials, and solid waste management [48].

In a study by Xue et al. (2015), the generation of furan, acids and water derived from red oak boosted by the incorporation of HDPE as a co-feed [49]. Aboulkas et al. (2012) investigated that co-pyrolysis of Tarfaya oil shale with waste plastics yielded higher bio-oil content in compare to pyrolysis of oil shale [50]. Oil derived from co-pyrolysis of oil shale and HDPE was found to have similar characteristics to that of transportation fuel [51]. Zang et al. (2021) observed that tar content and

char yield was increased by to the synergistic effect of pre-de-chlorinated Polyvinyl-chloride (DPVC) and Pingshuo (PS) coal during co-pyrolysis and there was significant decrease in yield of gas and pyrolytic water [52]. According to a study by Ahmaruzzaman et al. [53], complexity and molecular weight of compounds of bio-oil was decreased in case of co-pyrolysis of plastic waste with residual fuel oil (RFO). Similarly, it was estimated that the calorific value of char derived from biomass and plastics based co-pyrolysis have higher heating value than the char of individual pyrolysis of biomass [54]. Table 1 describes some recent studies on co-pyrolysis of biomass and plastics.

Many studies have shown that the quality of the bio-oil derived from the co-pyrolysis of biomass and plastics is enhanced due to the occurrence of synergistic effect during the reaction. The bio-oil resulted from the biomass and plastic wastes co-pyrolysis have higher heating value in comparison to bio-oil derived from the pyrolysis of individual feedstock [34,57,58]. According to Uzojeinwa et al. (2018), co-pyrolysis method is simple, cost-effective, and prominent method to produce high-quality biofuel [15]. The primary reason for the success of this method is the synergistic effect arising from the interactive reaction mechanism of the biomass and plastics during the process [72].

2.2. Catalytic co-pyrolysis's mechanism

Catalytic co-pyrolysis is defined as the process in which there is co-processing of different types of feedstocks in a pyrolysis process in presence of a catalyst. The incorporation of catalyst during the co-pyrolysis process enhances the overall efficiency of the pyrolysis reactions, potentially improving product yields and modifying the composition of the resulting bio-oil, bio-char and NCGs (non-condensable gases). It also reduces the coke production during the reaction [15]. Reaction mechanism for catalytic pyrolysis of biomass as shown in Fig. 1 highlights the combination of multiple reaction mechanisms involved in the conversion of biomass to desired products. Further, the catalyst has been shown to promote the pyrolysis reactions, leading to increased yields of desired products and or/reduction in unwanted byproducts [14]. Moreover, the addition of catalyst leads to the reduction in activation energy which further reduce the overall energy requirement of

Table 1
List of recent studies on co-pyrolysis of biomass and waste plastics.

Biomass	Plastic Feedstock	Type of Reactor	Operating Temperature (°C)	Feedstock mixture ratio (Biomass: Plastic)	Bio-oil yield/wt %	Heating value (MJ/kg)	Reference
Palm Shell	Polystyrene (PS)	Stainless steel type tubular reactor	500	1:1	61.63	38.01	[55]
Straw (cotton)	PP	Fixed bed reactor	380–480	1:1	35.8	46.90	[56]
Wood chips	Polypropylene (block)	Fixed bed reactor	500	1:1	63.10	45	[57]
Potato skin	HDPE	Stainless steel reactor	500	1:1	39	45.61	[58]
Pine based residue	Waste plastics	Stainless steel reactor	400	1:1	53	45	[54]
Bamboo	PP	Fixed bed reactor	500	8:2	50.95	27.24	[59]
Oakwood	PP	Fixed bed reactor	500	8:2	55.15	32.13	[59]
Niger seeds	Polystyrene (PS)	Stainless steel reactor	550	2:1	57.94	41.42	[60]
Karanja seeds	PS	Stainless steel reactor	550	2:1	57.81	42.18	[60]
Bamboo	PS	Fixed bed reactor	500	8:2	50.17	28.22	[59]
Date seeds	Plastic wastes	Fixed bed reactor	400	3:7	62.57	43.47	[61]
Empty fruit bunch	Low density polyethylene (LDPE)	Fixed bed reactor	510	1:1	67.1	–	[62]
Sugarcane bagasse	HDPE	Fixed bed reactor	600	6:4	63.69	42.41	[63]
Mahua seeds	PS	Fixed bed reactor	550	8:2	38.87	45.89	[64]
Mahua seed	Waste thermocol	Fixed bed reactor	525	1:1	74.2	–	[65]
Cellulose	Low density polyethylene (LDPE)	Fixed bed reactor	550	1:1	66	37.8	[66]
Rice husk	Waste tire	Fixed bed reactor	450	1:1	52	33.6	[67]
Cassava peel	PS	Fixed bed reactor	446.4	2:3	42.71	40.60	[68]
Groundnut shell	PS	Semi-batch reactor	510	2:1	62	–	[69]
Rubber seed cake	PS	Semi-batch reactor	500	1:2	65.85	41.10	[70]
Beech wood	PS	Tubular reactor	500	1:1	84.70	37	[71]

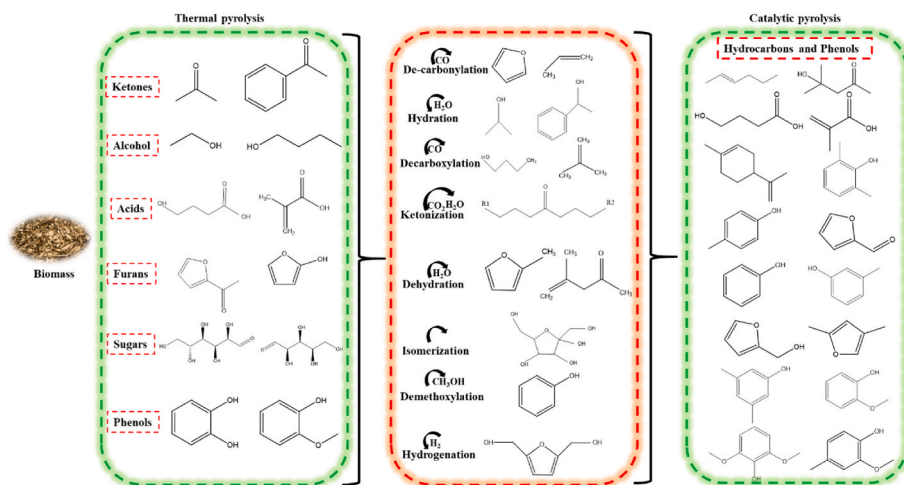


Fig. 1. Schematic representation of catalytic pyrolysis of biomass [14].

the process, thus increasing the total cost of the process [1].

Catalyst also increases the product selectivity which facilitates the requirement for the production of high value added products [73–75]. Similarly, the reaction mechanism for catalytic pyrolysis of LDPE over NiO and HY is presented in Fig. 2 which spotlights the catalyst selectivity towards the generation of aromatic compounds. The incorporation of HY boosted the Diels-Alder and other reactions due to which the generation of aromatic compounds increased [76].

Catalysts are utilized in co-pyrolysis to upgrade the bio-oil quality and yield [77]. In this process, the feed gets transformed into oil through thermal as well as catalytic cracking. The aromatic hydrocarbon production is enhanced during the catalyst based catalytic pyrolysis of biomass and waste plastics [34]. The linear olefins derived from waste plastics react with the furan compounds generated from biomass through Diels-Alder reactions, leading to the production of aromatic compounds as shown in Fig. 3 [78]. Consequently, Diels-Alder reaction

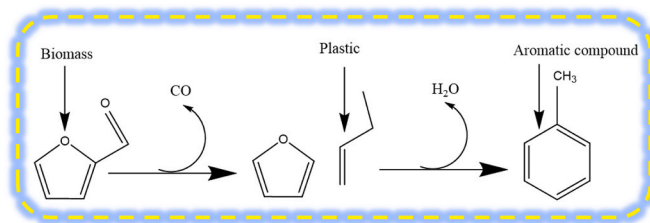


Fig. 3. Mechanism of Diels-Alder reaction.

followed by the dehydration reaction, increases the yield of aromatic compounds while simultaneously decreasing the coke production.

Fig. 4 depicts the reaction pathway for converting lignocellulosic biomass and plastics (LDPE) into aromatic products in the presence of

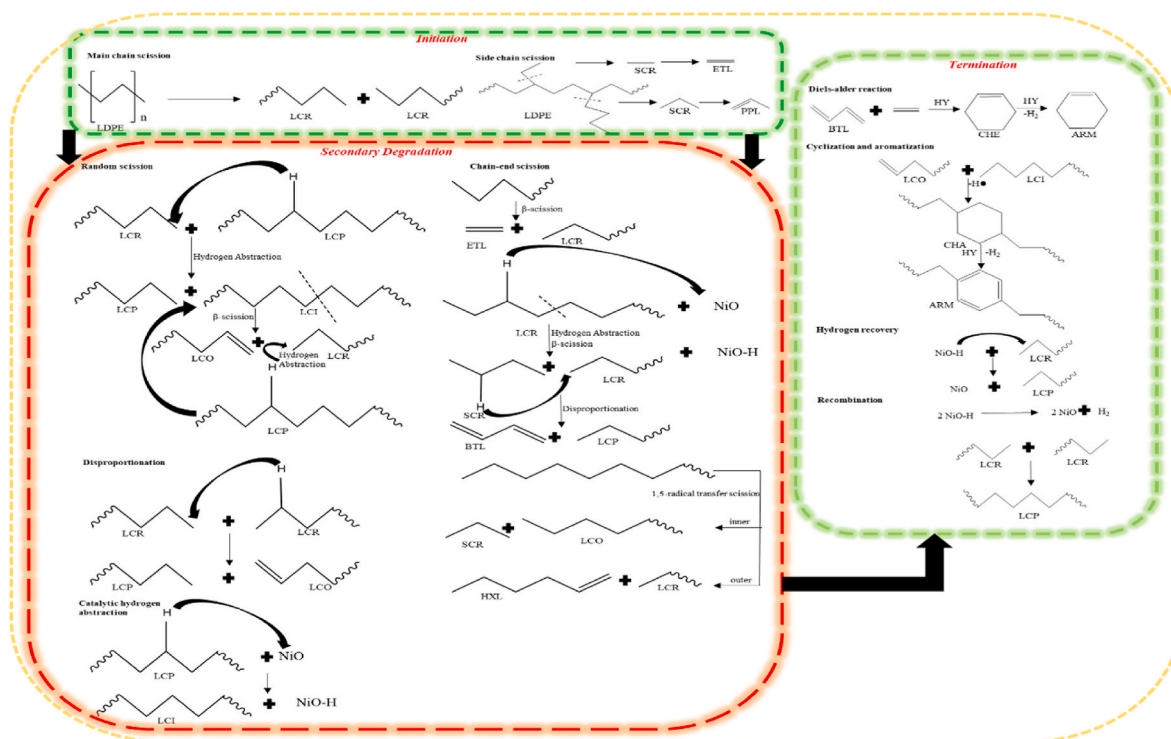


Fig. 2. Reaction mechanism of catalytic pyrolysis of LDPE over NiO and HY [76].

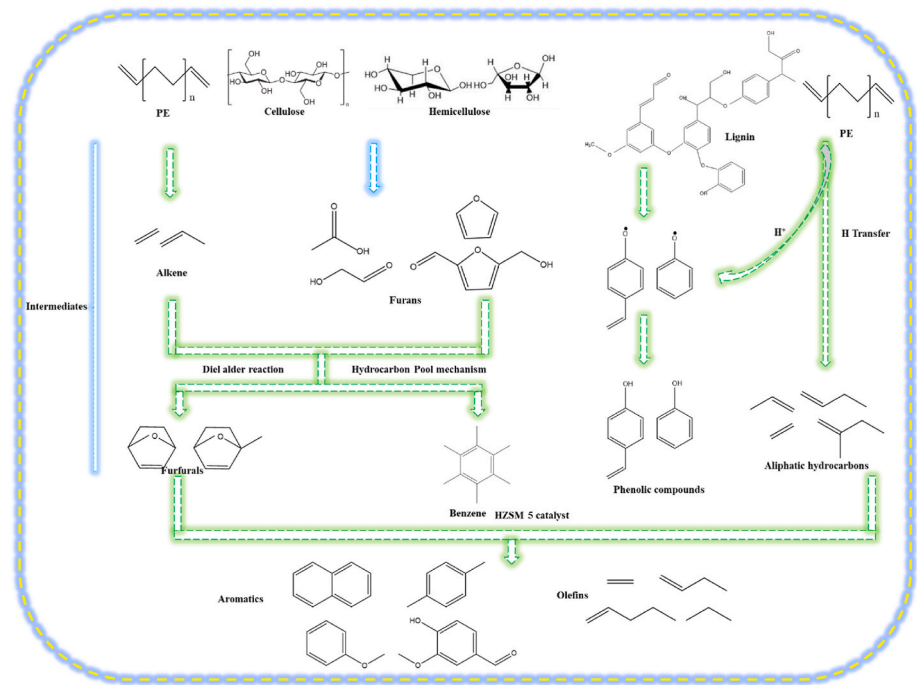


Fig. 4. Mechanism of reaction of catalytic co-pyrolysis of biomass and LDPE in presence of HZSM-5 catalyst [33].

HZSM-5 catalysts. When lignocellulosic biomass is thermally broken down, a succession of dehydration, decarbonylation, and decarboxylation processes yield furan compounds. Plastic hydrocarbons (e.g., olefins and alkanes) act as a hydrogen donor for cellulose based oxygenates, due to which coke formation in the catalyzed conversions gets reduced. The catalyst also promotes the cross-reactions by providing a surface for the molecules to react on. In the end, the products of the co-pyrolysis reaction desorb from the catalyst surface and are collected for further processing. Catalysts such as natural zeolite, Cu/Al₂O₃, Co/Mo/Z, Fe₂O₃ and Na-CaO-C, ZSM-5, Y-zeolite etc. are utilized in catalytic co-pyrolysis process [74,79,80]. Employing a variety of catalysts can enhance the synergistic transformation of these reactions, leading to the synthesis of high-value and innovative products, such as commercially viable chemicals, advanced materials [81]. Some recent studies on catalytic

pyrolysis and catalytic co-pyrolysis is illustrated in Table 2.

Yu et al. (2020) found fewer solid residues formation and lower content of acids, oxygenated compounds and nitrogen in non-catalytic co-pyrolysis whereas the co-pyrolysis of waste plastics (HDPE) and seaweeds using a catalyst boosted the concentration of aromatics and light hydrocarbons [86]. According to the study of Zhao et al. (2018), catalytic co-pyrolysis of lignite and straw produced prominent catalytic cracking effect on heavier compounds in comparison to non-catalytic co-pyrolysis [92]. Table 3 represents some recent studies based on the catalytic co-pyrolysis of biomass and plastics.

From the literature, it was observed that HZSM-5 is extensively used in the biomass (Pine based sawdust, terrified cellulose, etc.) and plastics (PP, HDPE, LDPE, etc.) co-pyrolysis. The production of aromatics is enhanced by the incorporation of HZSM-5 in the co-pyrolysis process.

Table 2
Literature survey of recent catalytic pyrolysis and catalytic co-pyrolysis studies.

Catalyst	Characteristics	Feedstock	Conditions	Bio-oil yield	Bio-oil composition	References
Fe-activated carbon catalyst	Specific surface area = 593.74 m ² /g	Pine sawdust	600 °C for 10 min	21.50 %	90 % phenolic content	[82]
WO _x -TiO ₂ -Al ₂ O ₃	Particle size = 0.2–0.3 mm	Organosolv lignin	600 °C for 30 min	36.2 %	1.6 % aromatics	[83]
HZSM-5	Specific surface area = 327.50 m ² /g, volume of pore = 0.178 cm ³ /g	Bamboo, PE	800 °C for 20 s	45.99 %	4.59 × 10 ⁸ a.u./mg BTX, 2.36 × 10 ⁸ a.u./mg total aromatics	[84]
ZSM-5	SiO ₂ /Al ₂ O ₃ ratio = 25	Cellulose, LDPE	450–750 °C for 60 s	55.6 %	46.5 % aromatics, 5.63 % olefins	[85]
SiC foam-MCM41	Specific surface area = 17.623 m ² /g, pore volume = 0.206 cm ³ /g	Tallow kernel oil	450 °C for 20 min	72.77 %	46.821 % aromatic compounds	[86]
Cu/MAS-C	BET surface area (m ² /g) = 120.31, pore volume (cm ³ /g) = 0.091, mean pore size = 3 mm	Melia azedarach sawdust	550 °C, 60 min	50.6 %	17.12 % phenolic content	[87]
Al-SBA-15	SiO ₂ /Al ₂ O ₃ = 40, pore size = 10.7 mm, surface area = 429 cm ² /g, micropore volume = 0.02 cm ³ /g	Xylan, LLDPE	413 °C for 10 min	59.24 %	12 % aromatics	[88]
MZSM-5	SiO ₂ /Al ₂ O ₃ = 40, pore size = 4.1 mm, surface area = 371 cm ² /g, micropore volume = 0.07 cm ³ /g	Avicel, LLDPE	394 °C for 10 min	57.99 %	24 % aromatics	[88]
FAU-SL	pore diameter = 25.51 mm	OPMF biomass	550 °C for 15 min	47 %	48.02 % acid rich carbonyls, 14.57 % aromatic compounds	[89]
MgAl-LDO/ZSM-5	Specific area = 334.3 m ² /g	Cyanobacteria	823 K for 30 min	41.1 %	45.314 % aromatic compounds	[90]
FA-ZSM-5	surface area = 329 m ² /g surface area, pore diameter = 37.18 Å	Pinnata waste	500 °C for 60 min	43.1 %	82.7 % aromatics compounds	[91]

Table 3

Overview of various studies on catalytic co-pyrolysis of biomass and waste plastics.

Biomass	Co-feedstock	Catalyst	Reactor	Operating temperature (°C)	Summary	Reference
Pine sawdust	LDPE	HZSM 5	Fixed bed reactor	450	Coking was prevented due to addition of LDPE and overall volatile content enhanced	[93]
Sugarcane bagasse	Polyethylene terephthalate (PET)	HZSM 5/Na ₂ CO ₃ /γ-Al ₂ O ₃	Tandem reactor	700	18.3 % BTX yield was obtained	[94]
Pine sawdust	HDPE	MgCl ₂ , HZSM 5	Fixed bed reactor	600	Overall hydrocarbon yield was enhanced	[95]
Pine	HDPE	ZSM 5	Fixed bed reactor	500	High bio-oil yield around 30.45 % with a gasoline selectivity of 99 %	[96]
Yellow poplar (Torrefied)	HDPE	Al-MCM-41, HZSM-5	Tandem microreactor	500	BTEX formation in Al-MCM-41 and HZSM-5 are 2.3 wt% and 24.6 wt% respectively	[97]
Torrefied cellulose	Polypropylene (PP)	HZSM-5, HBeta	Fixed bed reactor	550	BTEX formation in HZSM-5 and HBeta are 24.5 wt % and 9.0 wt% respectively	[98]
Sugarcane bagasse	HDPE	Faujasite type catalyst, FAU-EAFS	Fixed bed reactor	500	Bio-oil yield was around 68.6 wt% which have 74.6 hydrocarbon yield	[63]
Wood particle	Polypropylene (PP)	HBeta, HY zeolite	Fixed bed reactor	500	The bio-oil yields in HY and HBeta catalysts are 61.0 wt% and 60.5 wt% respectively	[99]
Jatropha oil	HDPE	ZSM5	Fixed bed tubular reactor	450	The liquid yield was around 68.03 wt% and yield of char was around 0.51 wt%. Waxy liquid formation was increased.	[100]
Jatropha oil	HDPE	ZSM5+SiAl	Fixed bed tubular reactor	450	The yield of liquid was around 54.20 wt% and biochar yield was around 0.13 %.	[100]

HZSM-5: Hydrogen form Zeolite Socony Mobil-5; Na₂CO₃: Sodium carbonate; γ-Al₂O₃: Gamma type Aluminium oxide; BTX: Benzene-Toulene-Xylene; Aluminium doped Mobil Composition of Matter No.41; MgCl₂: Magnesium chloride; FAU-EAFS: Faujasite type zeolite derived from electric arc furnace slag; HY zeolite: Hydrodeoxygenated zeolite.

Dorado et al. (2014) investigated that the yield of aromatics was enhanced in the co-pyrolysis of cellulose and polyethylene by the incorporation of HZSM-5 [101].

3. Modelling of co-pyrolysis process

Modelling of co-pyrolysis process plays a vital role in understanding and optimizing thermal decomposition reactions. There are some conventional models that are utilized for the kinetic analysis of pyrolysis process as shown in Table 4. These models include Ozawa Flynn Wall (OFW), Starink method, Friedman etc. There are different mathematical approximations which are assumed in these models. The objective of application of these models is to know about the thermal conversion behaviour of the feedstock. Thermogravimetric analysis (TGA) has proven to be a valuable technique to study the thermal degradation behaviour of various feedstocks [102,103]. According to Biswas et al. [104], the kinetic parameters derived by the TGA analysis can be utilized in process engineering for reactor and process design, modelling, and various industrial applications.

Activation energy can be calculated by using these conventional

models, after that pre-exponential factor(A), Gibbs free energy (ΔG), enthalpy (ΔH) and entropy (ΔS) can be determined by the given equations-

$$A = \beta E \cdot \exp\left(\frac{E}{RT_m}\right) / (R \cdot T_m^2) \quad (1)$$

$$\Delta H = E - RT_\alpha \quad (2)$$

$$\Delta G = E + R \cdot T_m \cdot \ln\left(\frac{K_B T_m}{hA}\right) \quad (3)$$

$$\Delta S = \frac{\Delta H - \Delta G}{T_m} \quad (4)$$

Here, T_m represents the peak temperature of DTG (Differential thermo-gravimetric) curve, K_B Boltzmann constant (1.381 * 10⁻²³ JK⁻¹), T_α the temperature at which the degree of conversion is α and h Planck constant (6.626 * 10⁻³⁴Js).

The reaction of solid state pyrolysis is predicted by plotting Criado z master plots using Eq. (5). This method is based on approximation of Senum and Yang which gives higher accuracy than 10⁻⁵% for x ≥ 20

Table 4

Conventional models used for the kinetic study of the co-pyrolysis process.

S No.	Model	Model's expression	Graph plots	Slope	Reference
1.	Ozawa, Flynn, Wall (OFW)	$\ln \beta = \ln \frac{A E_a}{R g(\alpha)} - 5.331 - 1.052 \frac{E_a}{R T}$	$\ln \beta$ vs 1/T	-1.052E _a /R	[7]
2.	Kissinger Akahira Sunose (KAS)	$\ln \frac{\beta}{T^2} = \ln \frac{A R}{E_a g(\alpha)} - \frac{E_a}{R T}$	$\ln (\beta/T^2)$ vs 1/T	-E _a /R	[105]
3.	Friedman method	$\ln \left(\beta \frac{d\alpha}{dt} \right) = \ln [A f(\alpha)] - \frac{E_a}{R T}$	$\ln \left(\beta \frac{d\alpha}{dt} \right)$ vs 1/T	-E _a /R	[106]
4.	Starink method	$\ln \left(\frac{\beta}{T^{1.92}} \right) = C_s - 1.0008 \frac{E_a}{R T}$	$\ln \left(\frac{\beta}{T^{1.92}} \right)$ vs 1/T	-1.0008 E/R	[106]
5.	Vyazovkin method	$I(E_a, T_a) = \int_0^{T_a} \exp\left(-\frac{E_a}{RT}\right) dT = \frac{E_a}{RT} p(x)$	-	-	[107]
6.	Coats-Redfern method	$\ln \left[\frac{g(\alpha)}{T^2} \right] = \ln \frac{A R}{\beta E_a} - \frac{E_a}{R T}$	$\ln \left[\frac{g(\alpha)}{T^2} \right]$ vs 1/T	-E _a /R	[107]
7.	Multicomponent distributed activation energy model (DAEM)	$\frac{d\alpha}{dt} = \int_0^\infty A_0 \exp\left(-\frac{E_a}{RT}\right) f(E_a) dE_a$	-	-	[107]
8.	Kissinger	$\ln \left(\frac{\beta}{T^2} \right) = \ln \left(\frac{A R}{E_a} \right) - \frac{E_a}{R T}$	$\ln \left(\frac{\beta}{T^2} \right)$ vs 1/T	-E _a /R	[108]

[106].

$$\frac{z(\alpha)}{z(0.5)} = \frac{f(\alpha) \cdot g(\alpha)}{f(0.5) \cdot g(0.5)} = \left(\frac{T_a}{T_{0.5}} \right)^2 \cdot \left(\frac{\frac{da}{dt} \alpha}{\frac{da}{dt} 0.5} \right) \quad (5)$$

Conventional models are easy to understand and apply however, these methods require the generation and analysis of thermogravimetric data on every application which is time consuming and require the availability of cost intensive equipment. Feedstock quality is the other factors which are not considered in conventional method to be mentioned. The machine learning is emerging as the potential tool that can remedy the issue of cost intensive equipment and the requirement of TGA analysis for each distinct feedstock [105,109]. Application of machine learning can develop a globalized model, is economically efficient and can be used for various purpose comprising of processes like optimization, modelling, process visualization, real-time analysis, scaling up of a process etc [1,110–112]. In order to overcome the drawbacks of conventional methods the application of machine learning models for the pyrolysis process is being focused in recent years.

4. Application of machine learning (ML) for the modelling optimization of co-pyrolysis process

Machine learning (ML) techniques are extensively being applied in the field of thermochemical conversion to enhance process optimization, product prediction, and overall efficiency. Machine learning (ML) has recently attracted more attention due to its ability to make prediction based on pre-published data and is also very much effective for the decision making from the data gathered from the laboratories and production sites [8]. For data classification and optimization of the overall process machine learning models are used, furthermore the machine learning application in a process eliminates expensive and time consuming experimental repetitions [113]. In comparison to conventional modelling and statistical techniques like thermodynamics equilibrium, computational fluid dynamics, and MATLAB etc., data-driven machine learning (ML) have shown good results [114]. Thus, machine learning models are very effective in processing with complex data sets in compare to traditional statistical tools. Nonlinear input-output relationships are mapped very effectively through the ML algorithms [30, 115]. In recent years, researches have investigated machine learning (ML) as an essential tool in the renewable energy sector for the modelling of reaction kinetics of thermochemical based conversion, process optimization, process control, and yield prediction [116].

Machine learning focuses on development of algorithms and make data based predictions. Machine learning (ML) methods have demonstrated significant potential in modelling thermo-chemical processes, as

well as forecasting and optimizing the yields and qualities of the resulting end-products. In general, there are four main types of machine learning algorithms: (a) *Supervised learning*, (b) *Unsupervised learning*, (c) *Reinforcement learning*, and (d) *Semi-supervised learning* as illustrated in Fig. 5 [117].

In supervised learning, the training and validation of the model are done by using the labelled input and output data sets. Supervised machine learning is basically classified into regression and classification methods [118]. In case of unsupervised machine learning, unlabelled data is analysed in order to detect any possible hidden patterns without predefined outcomes. Unsupervised machine learning attempts to analyse and find hidden patterns or information in unlabelled data. Clustering and dimensionality reduction are two subparts of unsupervised machine learning. A set of data points is categorized into groups or clusters. The clusters are formed on the basis of similarity and/or differences. The dimensionality reduction is used to convert training data from high to low dimensionality while containing the data variation [119]. In reinforcement learning, an agent (RL algorithm) learns how to make a series of choices in given environment to maximize the cumulative reward [120]. The labelled and unlabelled datasets are integrated by the semi-supervised learning during the training period. In case of time-consuming and complex process, the labelled data of supervised learning is supported by the semi-supervised learning [117]. Moreover, Explainable AI (XAI) is the advance process to provide understandable and interpretable explanations of decisions of ML models. XAI can be applied for the process of biomass and plastics co-pyrolysis for the explanation of the predictive modelling of products by ML models.

The complexity of co-pyrolysis of biomass and plastic waste necessitates the development of predictive algorithms to enhance the understanding and scalability of the process. Traditionally, kinetics-based models and computational fluid dynamics have been employed for this purpose, but they are computationally intensive. Alternatively, ML techniques offer a promising solution due to their efficiency and effectiveness in modelling. Fig. 6 depicts the ML approach for the prediction of yield (%) of bio-oi, bio-char and gases resulted from co-pyrolysis of biomass and plastics wastes. The advantage of ML based approach includes avoiding assumptions that could lead to inaccurate predictions, not requiring specific reactor design parameters, offering low computational cost.

4.1. Machine learning workflow

The typical workflow of the machine learning process is as follows (Fig. 7):

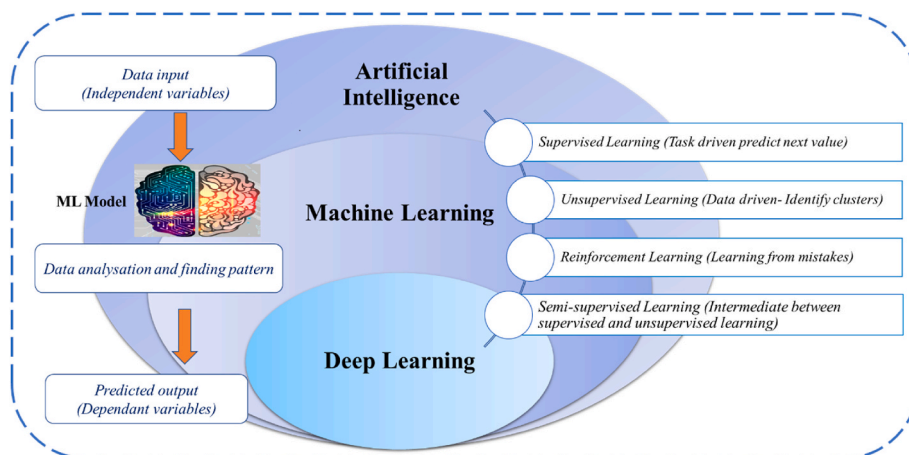


Fig. 5. Classification of artificial intelligence.

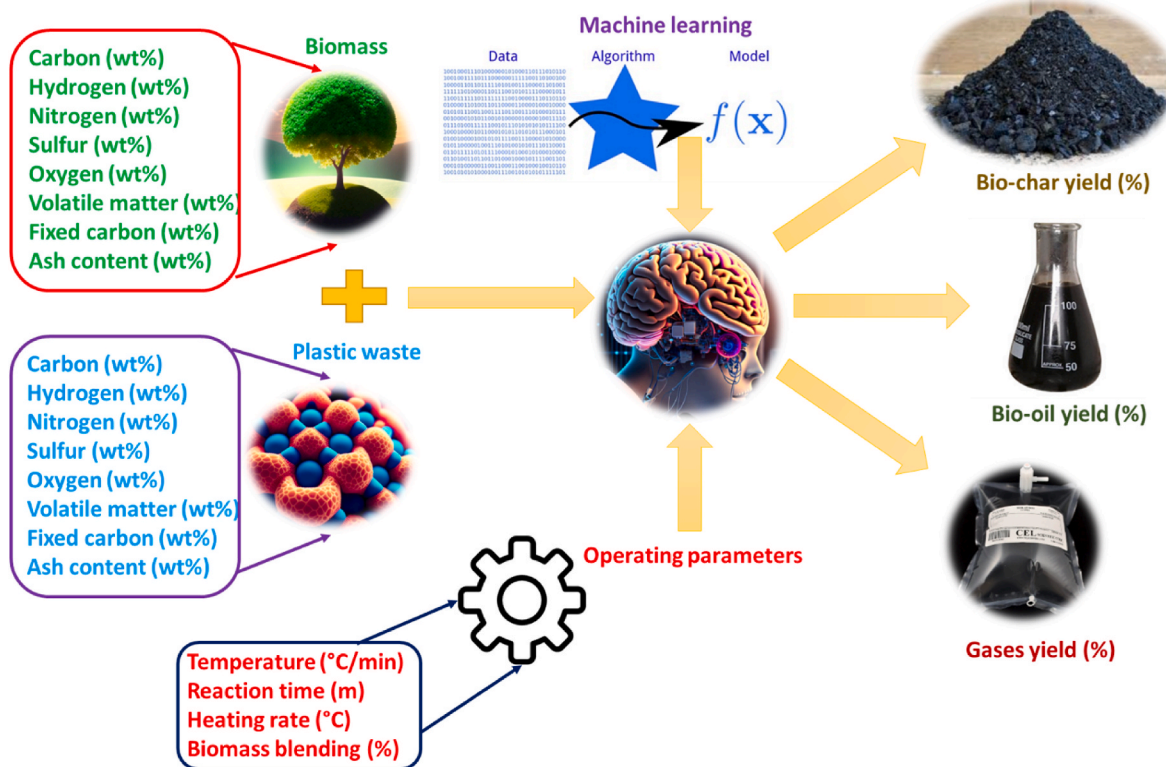


Fig. 6. Research process for prediction of yield (%) of products derived from co-pyrolysis of biomass and plastic waste.

- Identification of problem and goal: Defining the problem and the objective based on the process requirements and desired outcome
- Dataset collection: By thorough review of literature, screening the relevant dataset.
- Pre-processing of data: Pre-processing is done by removing the duplicates, handling missing values and addressing outliers using the effective ML tools/techniques.
- Splitting of dataset: Dividing the data points from dataset for training, validation, and testing in a prerequisite ratio (generally, 60:40, 70:30, or 80:20).
- Selection of algorithms/models: Based on the available and/or collected data, defining whether its classification, regression, clustering or any other type of problem which concomitantly helps select/choose the algorithm.
- Model training: ML model is trained by the training data.
- Hyper-parameter optimization: The optimal configurations are found by the adjustment of hyper-parameter of the model to give best performance.
- Model evaluation: Model's accuracy and dependability is evaluated by test data and assessed using different metrics such as coefficient of determination, mean percentage error, coefficient of determination (R^2), mean absolute error, absolute error, standard deviation, root mean standard error, etc. [120].
- Model deployment: After successful model evaluation and hyper-parameter optimization, the model is ready to solve problems. The comparison of the outcome with predicted result gives the effectiveness of the model.

Some part of the workflow is comprehended in the following subsection for better comprehensibility of the machine learning techniques.

4.2. Dataset collection

The necessary data for developing ML models for co-pyrolysis of

biomass and plastics can be gathered from suitable databases. Relevant articles can be selected for further processing by employing initial screening. The comprehensive coverage of the parameters essential for the ML model development can be provided by the literature data. Specifically, ML models are developed to predict output parameters such as yield of bio-oil, bio-char and syngas. The input parameters that can be employed are ultimate and proximate analysis of both biomass and plastic waste. Additionally, operating parameters such as biomass to plastic ratio, heating rate, reaction temperature, reaction time, feed to catalyst ratio, and catalyst characteristics (e.g., porosity, specific surface area and acidity) may be incorporated for prediction modelling [121].

Choosing appropriate input parameters that accurately capture the underlying physics of the process poses a significant challenge in machine learning (ML) modelling. This challenge is further compounded during modelling of co-pyrolysis process due to the diverse nature of biomass feedstock and plastic waste.

4.3. Pre-processing of data

Feature reduction or dimensionality reduction, is used to reduce the number of input variables (features) used in a model while retaining the most important or relevant ones. Feature reduction helps simplify the models, reduce computational complexity, improve model generalization, and increase the model accuracy. Principal component analysis and Pearson analysis are commonly used feature reduction techniques.

4.3.1. Principal component analysis (PCA)

PCA is one of the unsupervised machine learning tool used for reducing the dimensionality of data having a number of inter-related attributes (features or variables) while also retraining variation in the raw data. The PCA technique converts the data attributes to newer set of attributes which are often known as Principal Components (PCs). It maps every instance of the dataset in a d dimensional space to k

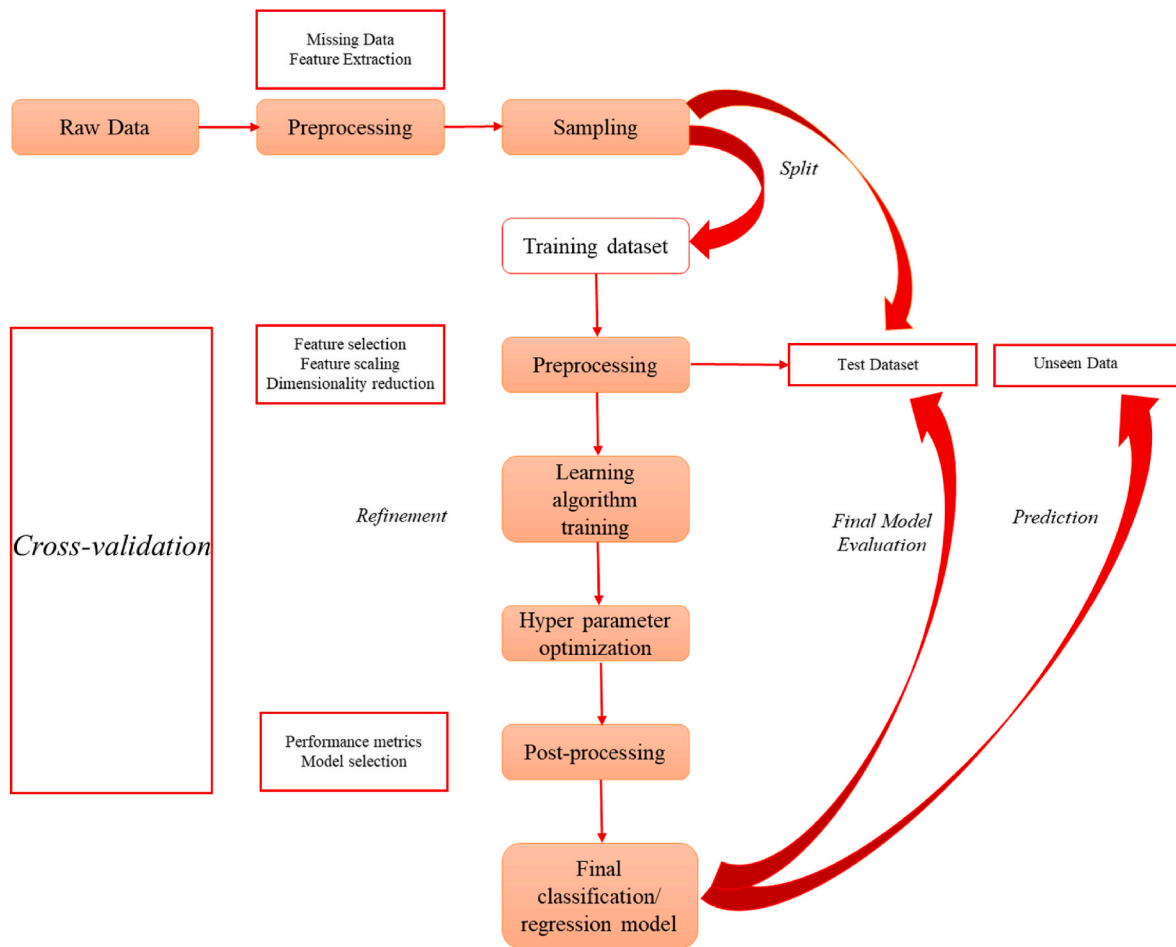


Fig. 7. Workflow of Machine Learning process.

subspace in such a way that $k > d$. The principal components are the newly generated set of k dimensions. Variance-covariance of a raw data in vector form is explored by PCA. In this analysis, the observations of correlated features are converted into a set of linearly uncorrelated features. PCA is utilized to visually represent the multidimensional distribution of the entire co-pyrolysis process [122]. Shahbeig et al. (2020) applied PCA to decrease the dimensionality of DTG (derivative thermo-gravimetric) data and to know the main reactions occurring in pyrolysis of biological wastes [123].

4.3.2. Pearson analysis

Pearson correlation coefficient (PCC) can identify the linear relationship of features of the established dataset [124], by using the equation-

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (6)$$

In the above equation, r is PCC, $\text{Cov}(X, Y)$ is variance between X and Y , and σ_X, σ_Y are the standard deviation of X and Y respectively. It is basically a two-tailed t-distribution with $(N-2)$ degree of freedom, which performs the significant level test as could be obtained in the following equation-

$$t = \frac{r\sqrt{(N-2)}}{\sqrt{1-r^2}} \quad (7)$$

Here, t stands for a statistical value and N represents number of total samples in the dataset. After applying the above equation, the p-value (significant level indicator) could be obtained. If the p-value is

acceptable, then PCC find out the linear correlations between any two variables. In case if p value is near to 1, then there will be higher possibility of existence of linear relationship. Potnuri et al. (2023) investigated that the feedstock properties and the operating parameters significantly impact the overall yield of bio-oil [125]. Pearson correlation coefficient matrix in Fig. 8 indicates that the properties of the feedstock and the experimental operating conditions have a significant effect on the yield of pyrolysis product. The strong positive correlations between feedstock properties and pyrolytic products yields demonstrate that the selection of feedstock is crucial in determining the yields of bio-oil, bio-char, and bio-gas. Additionally, operating conditions such as heating rate, temperature, time, and power significantly influence the yields of pyrolytic products.

4.4. Machine learning models

In this section of the review paper, discussion will be focussed on the different machine-learning methods used in biomass and plastic wastes co-pyrolysis. Different machine learning methods commonly employed in pyrolysis of biomass and plastic wastes comprise support vector machine (SVM), artificial neural network, etc. as depicted in Fig. 9 [116, 126,127]. The in-depth study of ML algorithms highlighted are elaborated as follows:

4.4.1. Artificial neural network (ANN)

An artificial neural network (ANN) is a class of machine learning models that is mimics the structure and role of brain of human. ANNs are composed of interconnected processing nodes, called neurons, that has memory to retain the information. Artificial neural network (ANN)

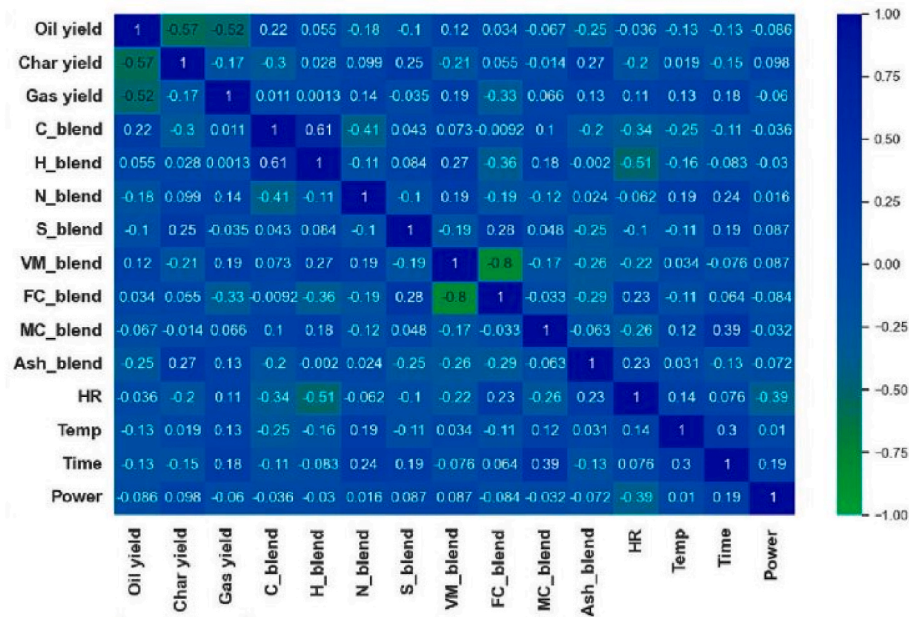


Fig. 8. Pearson correlation matrix for a pyrolysis process [125].

models are generally employed to correlate process-based parameters to the output received (Ismail et al., 2019).

The input layer of an ANN receives the raw data, and each subsequent layer processes the output of the previous layer. The output layer of the ANN produces the final output, which is used for prediction or classification. Through utilizing initial data, ANN establishes a link between the process's input and output parameters to provide the outcome. Kartal et al. (2022) used ANN to estimate activation energy during the pyrolysis of biomass [128]. Ullah et al. (2023) predicted the bio-oil obtained from the pyrolysis of micro-algae using ANN models with a R^2 value of 0.92 [129].

4.4.2. Support vector machine (SVM)

It is one of the machine learning approach that can be used to solve a variety of classification and regression problems [130]. It creates the ideal hyperplane for separating data from different datasets [124]. Data overfitting is less likely to occur with SVM models.

By using the kernel function in the SVM problem, SVM explains the correlation between the input and output variables. The kernel function is especially helpful in complex and non-linear problems.

One of the advantages of SVMs is that they are less susceptible to overfitting. Additionally, SVM can function well even with small datasets because it has a low computing cost and a low generalization error. Tang et al. (2020) applied SVM to forecast the yield of bio-oil and hydrogen with R^2 value higher than 0.85 [131]. Chen et al. (2018) forecasted char yield derived from biomass pyrolysis by application of SVM algorithm with R^2 value of 0.961 [132].

4.4.3. Decision tree

It is a kind of mathematical tool to solve both regression and classification problems. DT algorithm divides or splits data as shown in Fig. 9(a).

The most important property that divides the data into two or more homogeneous subsets is used to recursively partition the data to create the decision tree. The splitting process continues until a stopping requirement is met, such as when there are no further advancements in classification accuracy or when there is a minimum quantity of samples at each node. The data divides into tiny and sub-tiny portions to form a tree-like structure [133].

Three different kinds of nodes make the DT algorithm's tree like

structure: top node, leaf node, and the decision node. The decision tree can be utilized to address regression and classification problems. A decision tree in a classification problem predicts the class label of the incoming data using the decision rules it has learned from the training data. The decision tree forecasts a continuous value in a regression problem based on the mean of the target values at the leaf nodes.

Decision trees can be used for feature selection and can handle both continuous and categorical data. The risk of overfitting with decision trees increases with tree depth and noisy training data. Ullah et al. (2023) predicted yield of bio-oil from microalgae pyrolysis with R^2 value of 0.91 [129]. Xia et al. (2021) used DT algorithm to find the influence of input parameters in the biomass pyrolysis [8].

4.4.4. Random forest (RF)

RF is a statistical tool that helps to address the regression and classification problems [116]. Generally, the basic component of RF is DT. RF includes bundles of decision trees [134].

RF has a collection of decision trees, each of which based on a randomly selected of features and data. The RF's prediction is then generated by merging the results of all the distinct trees. By doing this, the likelihood of overfitting is decreased and prediction accuracy is increased [135].

RF has an advantage over other machine learning models since it can handle large datasets with plenty of features. It is also relatively easy to use and can provide insights into the key elements for making predictions. Tang et al. forecasted the yield of non-condensable gases (NCGs) derived from pyrolysis of biomass by RF model with R^2 value of R^2 greater than 0.85 [127]. Chen et al. (2020) applied RF model in the yield prediction of bio-oil and hydrogen content resulted by pyrolysis of biomass with larger feasibility [131].

4.4.5. Extreme gradient boosting (XGBoost)

XGBoost is one the machine learning model which uses a set of decision trees to generate predictions as shown in Fig. 9(c). It is an extensible gradient boosting tree method that has the potential to effectively solve the sparse data issue.

Decision trees are added to a model as part of XGBoost, with each tree fixing the mistakes produced by the one before it. This process continues until the desired number of trees is generated or until the model's performance plateaus. The algorithm also includes several

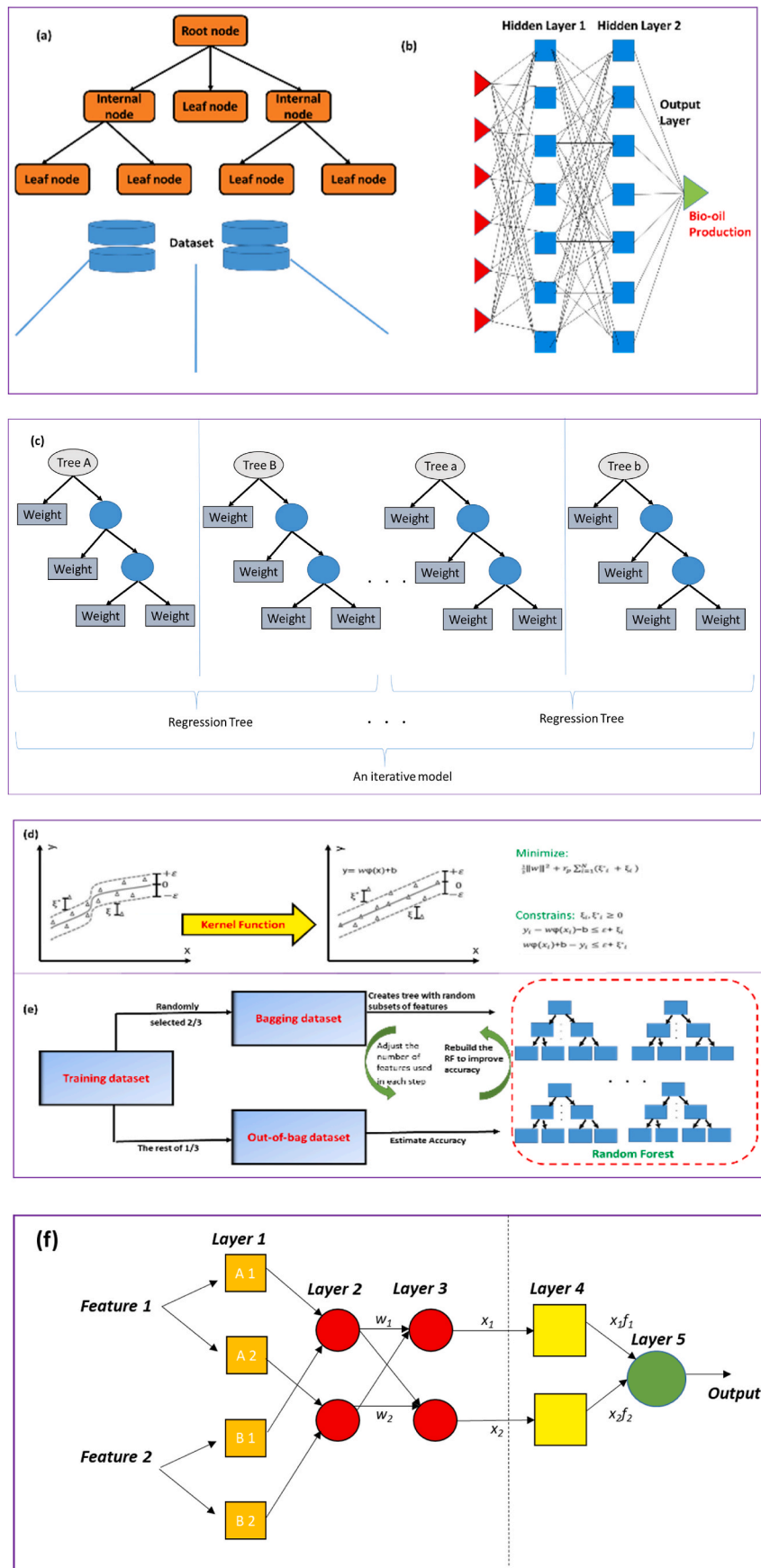


Fig. 9. Schematic of relevant ML methods. Fig. (a) Decision Tree (DT) (b) Artificial Neural Network (ANN) (c) Extreme Gradient Boosting (XGBoost) (d) Support Vector Machine (SVM) (e) Random Forest (RF) (f) Adaptive-network-based fuzzy inference system (ANFIS).

regularization techniques to prevent overfitting and improve generalization.

One of the strengths of XGBoost is its scalability, which makes it suitable for handling large datasets with many features [136]. It is also known for its speed and efficiency, allowing it to handle real-time prediction tasks. Gautam et al. (2022) used XGBoost to predict the yield of biochar with R^2 value of 0.97 derived from co-pyrolysis of biomass and plastic wastes [137]. Belden et al. (2023) investigated that XGBoost forecasted the bio-oil yield (BOY) from plastic waste pyrolysis with highest accuracy [110].

4.4.6. Adaptive neuro-fuzzy based inference system (ANFIS)

ANFIS is one of the intelligent model that combines the best features of neural networks and fuzzy logic. ANFIS is used to model and estimate complex systems with imprecision and uncertainty.

ANFIS works by using a set of fuzzy rules to map inputs to outputs as presented in Fig. 9(f). These rules are constructed using fuzzy logic, which allows for the representation of imprecise and uncertain knowledge. The inputs to the system are first fuzzified, which means that they are mapped to fuzzy sets that represent the degree of membership of each input in different linguistic terms (e.g., “low”, “medium”, “high”) [138]. The fuzzy rules are then used to combine the fuzzified inputs to produce a fuzzy output, which is de-fuzzified to obtain a crisp output.

The parameters of the fuzzy rules are tuned using a hybrid learning algorithm that combines backpropagation (a neural network training algorithm) and least squares (a regression technique). This allows ANFIS to learn from data and adapt to changing environments. Its ability to handle uncertainty and imprecision makes it a powerful tool for modelling complex systems.

4.5. Hyper-parameter optimization in ML

Hyper-parameters are tuned with the aim of improving the performance of the model. These are the configurations or settings that control the overall functioning of ML model. It is one of the most important step in ML modelling process because it has significant influence on the model's robustness, performance and proficiency [139]. In hyper-parameter optimization, the hyper-parameters are identified for the specific machine learning algorithm. The range of each hyper-parameter is defined during the optimization process. Generally, the techniques used in hyper-parameter optimization are grid search and random search. Grid search involves specifying a set of hyper-parameter values to be tried exhaustively. It involves systematically investigate all possible combinations of hyperparameters within defined search space. In random search, hyperparameters are explored randomly instead of trying all possible combinations.

Hyperparameter optimization should be performed within a cross-validation framework. Cross-validation helps assess the generalization performance of different hyperparameter settings by splitting the dataset into multiple subsets for training and validation. Hyper parameter optimization integrates with a 5-fold cross-validation to mitigate overfitting and enhance the predictive capability of the model. Some critical hyper-parameters were optimized based on the past reported studies [122]. According to a study of Li et al. support vector model with optimized hyperparameters predicted the properties of biochar with R^2 value of 0.94 [140]. Alabdrabalnabi et al. (2022) got best accuracy of XGBoost model (RSME = 70) for the prediction of bio-char after the hyper-parameter optimization [38].

4.6. Model evaluation

Trained model performance is measured by different performance metrics such as Accuracy, Precision, F-Measure, and the area-under-the-curve (AUC). Coefficient of determination R^2 , the root mean square and the mean absolute error (MAE) are most commonly used metrics [141].

Coefficient of determination R^2 is used to evaluate the proportion of

the variance of the dependent variable that is predictable form the independent variables in the model as shown in Eq. (8):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

The experimental value and the predicted value of the target variable are y_i and $\hat{y}_i$ respectively. The mean of the observed target variables is represented by $\bar{y}$. When there is 100 % accuracy in the model, then the value of R^2 becomes one. Moreover, if the model produces poor predictions then value can also be negative. If R^2 value is 0 then it means model doesn't explain any variability in data and value above 0.5 is considered good.

Mean Absolute Error (MAE) measures the average absolute difference between the observed and predicted value as shown in Eq. (9):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9)$$

The root-mean-square error (RMSE), evaluates the square root of the average of the squared differences between the predicted values and the observed values in a dataset as shown in Eq. (10), the best fit model is considered when the value is equal to zero.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

4.7. Explainable Artificial Intelligence techniques (XAI)

Model accuracy, fairness, transparency and outcomes can be characterized by XAI [142]. Machine learning models can be integrated with Explainable Artificial Intelligence (XAI) to enhance transparency, accountability, and trust. Commonly used XAI's like Shapely Additive Explanations (SHAP) and Partial Dependence Plots (PDPs) are comprehended in the following sub-sections.

4.7.1. Shapely Additive Explanations (SHAP)

SHAP values are used for analysing the training features for prediction of model yield [38]. SHAP values facilitate the construction of input and output variables in a model [143]. SHAP values are calculated for individual features in ML modelling to understand their impact on the prediction of specific value. SHAP analysis helps to understand influence of input variables on bio-oil yield derived from co-pyrolysis of biomass and plastic wastes [144]. ML models are built on the basis of SHAP values to determine the impact and relative contribution of input to the predicted output variables. Alabdrabalnabi et al. (2022) determined the impact of features on the product yields prediction using SHAP analysis [38]. P. Prasertpong et al. (2023) used SHAP value and found that co-pyrolysis conditions based variables contributed 35 % for model prediction of yield of bio-oil [145]. Alabdrabalnabi et al. (2022) utilized SHAP technique to understand the impact caused by various operating parameters on the yield co-pyrolysis based bio-oil [111] as shown in Fig. 10.

4.7.2. Partial Dependence Plots (PDPs)

PDPs applied to understand the effect caused by the selected variable on the output of a machine learning model while keeping all other variable constant [146]. It serves as a tool for visually exploring the nature of dependence (whether linear, monotonic, or intricate) between output and input variables. PDPs can be generated for analysing the different machine learning models or algorithms e.g., XGBoost, SVM, etc. It helps to understand the importance of specific parameter within the context of the specific models, aiding in feature selection, model validation, and overall model interpretation. Zhu et al. (2019) found that PDP analysis was very effective for optimization of biomass

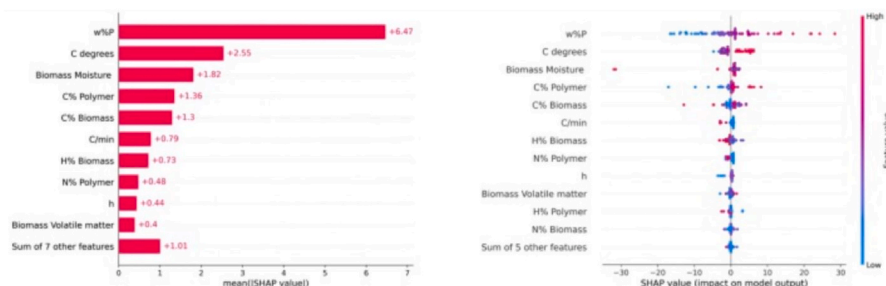


Fig. 10. SHAP analysis diagram for prediction of bio-oil from co-pyrolysis process [111].

pyrolysis and to understand the mechanism of reaction [147]. Potnuri et al. (2023) employed PDP to analyse the influence of variables on the yield of products derived from catalytic co-pyrolysis of biomass and plastic wastes [125]. Xu et al. (2023) used PDP analysis and found that elevating the temperature to 500 °C exhibited a predominant influence on bio-oil yield [122].

4.8. Literature survey of machine learning models in pyrolysis of biomass and plastics

ML models have been used to model the co-pyrolysis process as reported by many studies [148]. Table 5 gives the overview of ML approach in co-pyrolysis process of biomass and plastic waste. The most prominent ML method used to model the co-pyrolysis process is ANN [116]. Because of their ability to find out complex as well as non-linear relationships among dependant and independent variables, artificial neural networks (ANNs) are well-adapted for modelling the pyrolysis or co-pyrolysis process. ANNs have been applied in several co-pyrolysis-related studies, including the bio-oil yield prediction from different feedstocks, the characterization of different products, and the optimization of co-pyrolysis parameters for maximum yield or quality [149]. For example, models based on artificial neural networks (ANNs) are employed to predict various aspects such as pyrolytic conversion, yield of non-condensable gases, biochar yield, kinetic parameters and remaining mass [150]. Balsora et al. (2022) used four different models of ANN for the prediction of the different kinetic parameters of different biomass and the input variables of the analysis were TGA and primary analysis (proximate, ultimate and biochemical analysis) of biomass feedstocks [151]. The study suggests that performance of ANN 4 and ANN 3 was effective in prediction of the kinetic factors ($R^2 \sim 0.99$). Hough et al. (2017) found that the use of ANN and DT significantly reduced the computational expenses associated with a sophisticated kinetic model for lignin pyrolysis, making them feasible components for constructing robust models [112].

In sludge pyrolysis, the RF model was very accurate in prediction, and value of R^2 was higher than 0.813 [44]. Bio-oil yield (BOY) and its hydrogen content based on ML models was predicted by W. Zhang et al. (2021) [152]. A study comparing the approaches taken by RF and MLR was conducted. The results shown that RF models ($R^2 \sim 0.97$) are more accurate than MLR ($R^2 \sim 0.37$) to predict the bio-oil content and hydrogen content [153]. Zhang et al. (2019) employed RFR model for prediction of mass retained after the manure pyrolysis with R^2 value of 0.9996 [154]. Yang et al. (2022) predicted the bio-oil yield by applying the RF model with the prediction accuracy (R^2) of 0.925 [155]. Similarly, Hai et al. (2023) applied RF model to for the prediction of bio-char yield with R^2 value of 0.85 [156]. According to a study by Leng et al. (2021), R^2 values of RF model for predicting yield of pyrolyzed oil, char and gas are 0.9479, 0.9545 and 0.9284 respectively in pyrolysis of lignocellulosic biomass [157].

SVM has been used in pyrolysis research to model the effects of various input variables on the production of biofuels and other valuable products. For instance, SVM has been used to model the influence of

temperature and size of particle on the yields of different pyrolytic products. The models of ANN ($R^2 \sim 0.91$) and SVM ($R^2 \sim 0.89$) have predicted pyrolytic products and their heating value [116]. SVM models were employed to predict the fuel properties of bio-char, pyrolytic conversion, biological waste pyrolytic conditions, yield of bio-oil, product distribution [123,140,158]. For the prediction of the char yield derived from cattle manure pyrolysis, models based on Least Squares Support Vector Machines (LS-SVM) and ANN were utilized [159]. The correlation coefficients (R^2) for LS-SVM and ANN were 0.96 and 0.80, respectively, suggesting superior performance of LS-SVM.

XGBoost can be easily trained on experimental data to find out the most prominent factors that influence pyrolysis mechanism and to predict the different product yields. Prasertpong et al. (2023) employed XGBoost by using 26 input features for predicting bio-oil yield in biomass and waste plastics co-pyrolysis and got prediction accuracy (R^2) of 0.86–0.90 [18]. Alabdralnabi et al. (2022) used ML models for modelling the influence of temperature and rate of heating on bio-oil yield (BOY) and bio-char [111]. XGBoost and dense neural network (DNN) predicted the yields pyrolytic products [38].

Effectiveness of MLR depends on the complexity of the system and the primary relation among the input and output variables. Conag et al. (2019) applied MLR model for prediction of the HHV (MJ/kg) of raw and torrefied sugarcane residues with R^2 value of 0.90 [160]. Ullah et al. (2021) used MLR model to predict the yield of bio-oil derived from the biomass pyrolysis with R^2 value of around 0.75 [146]. Zhang et al. (2019) employed MLR model to predict mass remained after cattle manure pyrolysis with prediction accuracy of around 0.8583 [154].

In summary, the predominant machine learning (ML) model employed in the pyrolysis or co-pyrolysis process were observed to be ANN, followed by SVM, RF and DT. Input variables used for the modelling of pyrolysis process are heating rate, residence time, operating temperature, moisture and ash content, particle size of feedstock, fixed carbon, volatiles, ultimate and proximate analysis, mass of catalyst, feedstock ratio, flow rate, etc. The frequently examined output variables are product distribution, HHV of char and oil, mass loss, yield of bio-char as well as bio-oil, and mass % of carbon, nitrogen, oxygen and hydrogen. The application of ML modelling for yield prediction proves of reactor conditions.

4.9. Challenges of machine learning (ML) approach in co-pyrolysis process

More robust ML models are needed for more accurate prediction of biomass-plastic co-pyrolysis process. Achieving this goal requires compiling a more comprehensive database. The major cause of inability of ML modelling in co-pyrolysis. One of the major problem in ML modelling of catalytic and non-catalytic co-pyrolysis process is due to the lack of sufficient reported data [44]. Furthermore, a drawback of purely data-driven black-box machine learning models (ML) lies in their limited interpretability. While certain ML algorithm (such as ANN) and model explainers (like the SHAP method) can offer insight into feature importance and correlations with the predicted targets for co-pyrolysis,

Table 5

Literature survey of application of different models of Machine learning (ML) on pyrolysis of biomass and plastics.

Feedstock	ML Model	Input parameters	Output parameters	Data set size	R ²	Conclusion	References
Plastic: PE, PP, PS, PVC	ANN, GP, SVM, DT	Type of plastic, Ultimate analysis, Particle size, Feed intake, temperature, Vapour residence time	Product yield and its composition	274	ANN = 0.719 GP = 0.534 SVM = 0.653 DT = 0.86	The outstanding capability of the product prediction was carried out by the fitrensemble model of DT with ultimate analysis.	[37]
Waste sludge	RFR	Proximate and ultimate analysis, operating temperature, heating rate, time of reaction	Yield of bio-oil, yield of syngas, composition of syngas (H ₂ , CO, CO ₂ , and CH ₄ concentration), biochar yield, biochar atomic composition (i. e., O/C, H/C, and H/N ratio) biochar calorific value	182	R ² = 0.813-0.972	A remarkable prediction accuracy showed by the RFR model.	[113]
Biomass and plastics	XGBoost	Proximate analysis, Ultimate analysis, reaction temperature, feed ratio, sample mass loading and sample size	Bio-oil yield (BOY) and Synergistic effect (SE) of biomass and plastics	32	R ² = 0.85-0.95	The development of XGB models was successfully done.	[18]
Biomass and plastic	DNN, XGBoost	Ultimate, proximate, pyrolysis conditions and weight % of polymer	Bio-oil and bio-char yield	94	DNN = 0.96 XGBoost = 0.97	The method can be promising by utilizing SHAP with XGB and DNN for predicting the bio-oil and bio-char yield after incorporation of biochemical composition.	[38]
Biomass	SVR, RFR, GBR	Process conditions, ultimate analysis and proximate analysis	Product yield, yield of bio-oil, gas yield, gas composition (concentration of H ₂ , CO, CO ₂ , and CH ₄), char yield, biochar characteristics	249	SVR = 0.3-0.7 RFR = 0.7-0.91 GBR = 0.822-0.94	The most accurate and generalized prediction performance was given by GBR.	[127]
Biomass	LR, XGB, RF, SVM	Volatile matter (%), ash content (%), fixed carbon (%), ultimate composition	Biochar yield and HHV	240	XGB = 0.91 RF = 0.89 LR = 0.49 SVM = 0.63	The highest correlation coefficient was given by the XGB algorithm with optimized parameters	[161]
Biomass	ANN models	Proximate, ultimate, and biochemical analysis	Order of reaction, activation energy, and frequency factor	704	R ² = 0.99	ANN-4 and ANN-3 given the kinetic parameters which are in accordance relative to the kinetic determined by the analytical method	[151]
Microalgae	DT, SVM, GPR, ANN	Ultimate composition, proximate composition, and pyrolysis parameters	Bio-oil yield	1523	ANN = 0.8950 GPR = 0.95 ET = 0.72 SVM = 0.45 DT = 0.87	There was a significant impact on the bio-oil yield on output by the optimal parameters chosen by PSO	[129]
Lignocellulosic biomass	MLR, DT, RF, GB (gradient boosting)	Proximate analysis, ultimate analysis, and process conditions	Bio-oil yield	315	MLR = 0.26 DT (not pruned) = 0.80 DT (pruned) = 0.77 RF = 0.79 GB = 0.89	GB method showed the highest R ² value	[153]
Agricultural biomass	RF, MLR, DT, KNN	Proximate analysis, ultimate analysis, and pyrolysis conditions	Biochar yield	292	RF = 0.855 MLR = 0.821 DT = 0.559 KNN = 0.510	The higher accuracy of biochar yield prediction was shown by Random Forest	[156]
Agricultural biomass	RF, MLR, DT, KNN	Proximate analysis, ultimate analysis, and pyrolysis conditions	The specific surface area of biochar	292	RF = 0.804 DT = 0.618 KNN = 0.557	The higher accuracy of prediction of the specific area of the biochar was shown by Random Forest	[156]

(continued on next page)

Table 5 (continued)

Feedstock	ML Model	Input parameters	Output parameters	Data set size	R ²	Conclusion	References
					MLR = 0.292		
Rice husk and polystyrene	SVR	Blending ratio, heating rate	Water, bio-char and bio-oil yield, heating rate conversion (%)	11	0.81-0.94	R ² value for prediction was greater than 0.8	[162]
Tea waste and polystyrene	SVR	Proximate and ultimate analysis	Yield (%) of oil, gas and char	11	0.85-0.93	SVR showed high prediction accuracy than other models	[163]

ANN: Artificial neural network; GP: Gaussian process; SVM: Support vector machine; DT: Decision tree; RF: Random forest; XGB: XG boost; SVR: Support vector regression; DNN: Deep neural network; GBR: Gradient boosting regression; KNN: K nearest neighbour; MLR: Multiple linear regression; ET: Ensemble tree.

this information only indicates the extent to which input features impact prediction performance. Understanding why these input features exert such impacts still remains unclear.

5. Challenges and opportunities

Co-pyrolysis presents a promising technique for converting biomass and waste plastics into sustainable fuels and chemicals. However, the process faces several challenges, including variability in feedstock composition, catalyst instability and deactivation, and the complexity of synergistic interactions. Future research should focus on advancing catalytic co-pyrolysis to develop cost-effective catalysts capable of producing high-quality bio-oil with enhanced yield, and selectivity of the target chemicals. This approach would facilitate researchers to do large scale and impactful future researches. These insights could lead to optimized process designs and improved operational strategies for industrial applications.

Additionally, further research in this field should explore the integration of deep learning algorithms in co-pyrolysis of biomass and plastics for bio-fuel production to enable more precise predictions, control and optimization. Leveraging the robust predictive capabilities of machine learning models within comprehensive systems frameworks could offer significant benefits to stakeholders and entrepreneurs. These include improved profitability through enhanced productivity, precise prediction, and the automated control and optimization of the co-pyrolysis processes, leading to better quality assurance and production efficiency.

The advancement in the production and application of bio-based products has gained considerable attention as a sustainable solution to address rising energy demands. Moreover, their adoption reduces reliance on crude oil resources. Employing these products as biofuels offers a carbon-neutral alternative. Substitution of these products for conventional fossil fuels in energy generation, these biofuels contribute to carbon credit accrual. Additionally, the application of biochar as a soil amendment in agriculture promotes carbon sequestration. The generation of value-added products from co-pyrolysis of lignocellulosic biomass and plastic wastes further facilitates solid waste management, particularly for agricultural residues and waste plastic garbage. Thus, the co-pyrolysis of these feedstocks offers substantial environmental advantages while simultaneously supporting the energy sector.

6. Conclusion and future recommendation

The current research trend related to the catalytic as well as non-catalytic co-pyrolysis of different biomass and waste plastics is comprehended in the present review paper, wherein further the advancements in the application of machine learning, its advantages and challenges in terms of pyrolysis process are highlighted. Blending of plastic wastes with biomass increases overall effective ratio of H/C of overall feedstock due to which synergism occur during the process. Numerous researchers have acknowledged biomass and plastic wastes co-pyrolysis process as a substantial enhancement for both the quantity and quality of bio-oil. From the current research scenario, it was

observed that zeolite catalysts are the focus of multitude of research works due to their favourable properties (surface area, catalytic activity, selectivity, etc.), however present research trend is also shifting towards waste derived catalysts that are on par with the commercial catalyst in order to attain circular economy approach.

Further, application of machine learning is very effective to forecast yield of different products in biomass and plastic wastes co-pyrolysis using raw material characteristics, operating parameters and plastic content in feedstock as the input parameters for an input-output model. Machine learning approach in this process eliminates the time-consumption and hard experimental work. There are several ML models which have applied on different pyrolysis system to predict the yield of bio-oil, aromatics, phenolic content etc.

Similarly, several ML techniques like the hyperparameter optimization and XAI tools are also very effective for improved predictions of various pyrolysis products as well as to explain the effect of various process conditions on the co-pyrolysis process. However, there is lack of studies utilizing these tools to enhance the interpretability of the black box machine learning models. Further research and development in this area are needed to fully realize the potential of ML in co-pyrolysis.

Deeper understanding of the machine learning needs to be obtained in order to utilize various interactive tools which are found lacking in the current literature available. Similarly, integration of ML tools that optimize the working of the model like hyperparameter tuning, principal component analysis, SHAP, PDPs, etc., are need to be focussed on to increase the efficacy of the model as well as to reduce variances and biases which will help obtain robust and globalized models.

CRediT authorship contribution statement

Deepak Bhushan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Sanjeevani Hooda:** Visualization, Software, Formal analysis, Data curation. **Prasenjit Mondal:** Validation, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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